Figure 8: Passphrase has been changed

enabled with the key file supplied. Now Slot3 can also be used to decrypt the partition.

Restoring LUKS headers

For some commonly encountered LUKS issues, LUKS header backups can mean the difference between a simple administrative fix and permanently unrecoverable data. Therefore, administrators of LUKS encrypted volumes should engage in the good practice of routinely backing up their headers. In addition, they should be familiar with the procedures for restoring the headers from backup, should the need arise.

LUKS header backup

LUKS header backups are performed using the *cryptsetup* tool in conjunction with the *luksHeaderBackup* sub-command. The location of the header is specified with the *--header-backup-file* option. So by using the command given below we can create the backup of any LUKS header:

```
# cryptsetup luksHeaderBackup /dev/vdb1 --header-backup-file /
root/back
```

As with all systems administration tasks, LUKS header backup should be done before every administrative task performed on a LUKS-encrypted volume. Should the LUKS header be corrupted, LUKS stores a metadata header and key slots at the beginning of each encrypted device. Thus, corruption of the LUKS header can render the encrypted data inaccessible. If a backup of the corrupted LUKS header exists, the issue can be resolved by restoring the header from this backup.

Testing and recovering LUKS headers

If an encrypted volume's LUKS header has been backed up, the backups can be restored to the volume to overcome issues such as forgotten passwords or corrupted headers. If multiple backups exist for an encrypted volume, an administrator needs to identify the proper one to restore. The header can be restored using the following command:

```
# cryptsetup luksHeaderRestore /dev/vdb1 --header-backup-file
/root/backup
```

Now, let's suppose someone has changed the password of

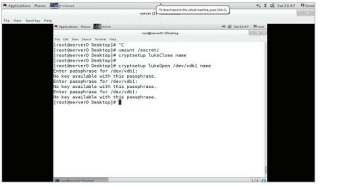

Figure 9: Decrypting a partition with the passphrase supplied initially

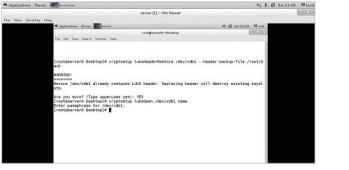

Figure 10: Header is restored from the backup file

the encrypted partition */dev/vdb1* using *luksChangeKey*, but the password is unknown. So the only option is to restore the partition from the backup that we have created above, so that we can decrypt the partition from the previous passphrase. The backup also helps when admin forgets the passphrase.

In Figure 8, a backup of */dev/vdb1* has been taken initially, and its passphrase has been subsequently changed by someone, without our knowledge.

Before closing a partition, we have to unmount the locked partition. After closing the partition, trying to open the partition by using the previously set passphrase will throw the error 'No key available with this passphrase', because the passphrase has been changed by someone (Figure 9).

But as the backup has already been taken by us, we just need to restore the LUKS header from the backup file which was created earlier. As shown in Figure 10, the header has been restored.

Now we can open the header with the passphrase that was set earlier. Therefore, it is always beneficial for administrators to create a backup of their header, so that they can restore it if somehow the existing header gets corrupted or a password is changed. End

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